Saltwater intrusion

If excessive water is taken from the aquifers along the coast, the sea water enters the coastal aquifer. This process is termed as saltwater intrusion.

5.4 Cryosphere

Cryosphere includes the water in frozen state. Glaciers, ice sheets, ice caps, lake and river ice, permafrost, seasonal snow and ice crystals in the atmosphere together form cryosphere. Earth's climate is highly influenced by the extent of cryosphere as it controls the energy budget of the earth (Figure 5.2).

Perennial ice cover is found in Greenland and Antarctica as ice sheets, as mountain glaciers and as permafrost in higher latitudes. **Permafrost** is the condition prevailing when water freezes above and below the ground, (including rock or soil) for more than two consecutive years. Most permafrost regions are located in high latitudes, but alpine permafrost may exist at high mountains in much lower latitudes.

Fact File

Mount Kilimanjaro (5895m) in Tanzania, Africa, located closer to the equator has permafrost.

Seasonal snow and ice crystals are confined to middle latitudes and high mountains in lower latitude. **Sea ice** is frozen ocean water. Its formation, growth and melting are all confined to the ocean. An **ice shelf** is a thick, floating slab of ice that forms where a glacier or ice flows

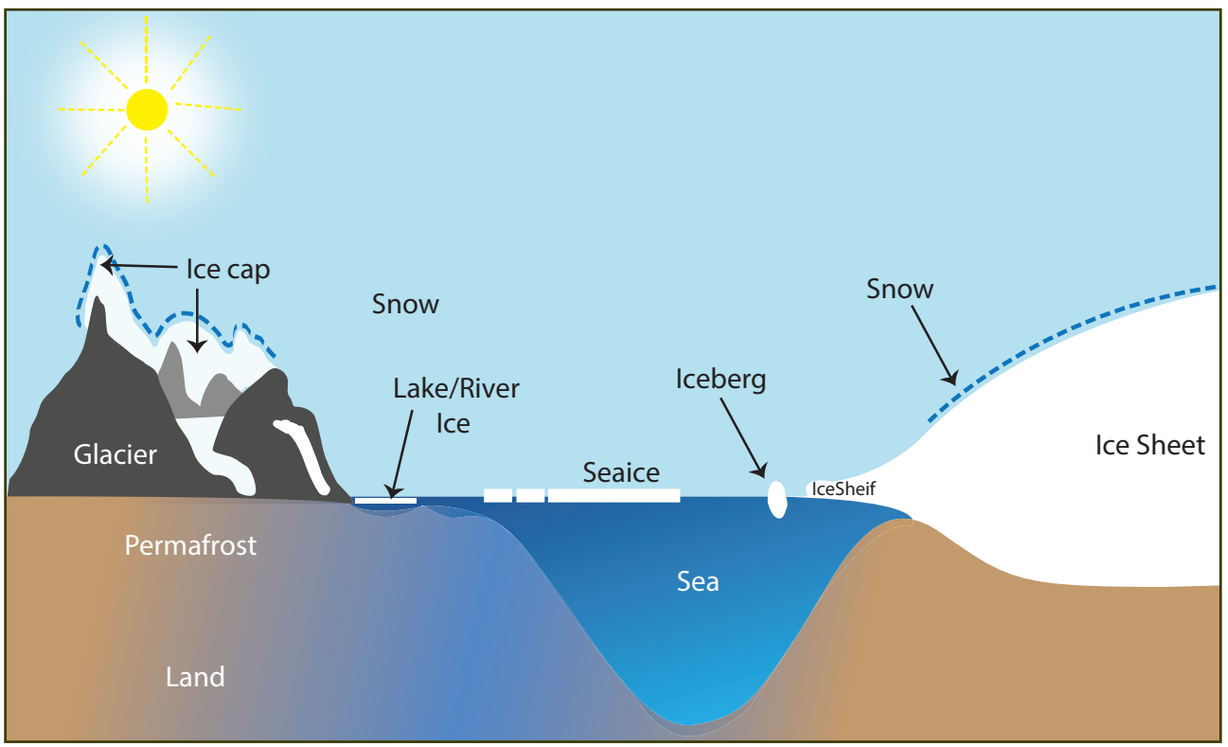


Figure 5.2 Cryosphere

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